

EXECUTIVE SUMMARY

This project evaluates how a healthcare organization could assess the value of adopting artificial intelligence for medical diagnostics by comparing three approaches: a human-status-quo model, a logistic regression model, and a deep learning model. The analysis goes beyond basic performance metrics by examining precision, recall, F1 score, error costs, net tangible benefit, comparative net tangible benefit, and scenario-based tradeoffs between patient experience and diagnostic sensitivity. Overall, the project finds that while simpler AI models can deliver major efficiency gains, the deep learning model provides the strongest overall value because it reduces costly false negatives, improves clinical accuracy, and performs best across multiple organizational priorities. The project also proposes a broader value-measurement framework that encompasses financial, operational, patient-centered, and equity considerations to support the responsible long-term adoption of AI in healthcare.

TASK 1: Understanding the approaches

1) Human Status Quo Model

Precision (TP/TP+FP)	$50/(150 + 450) = 0.25$ (25%)
Recall TP/(TP+FN)	$50/(150+50) = 0.75$ (75%)
F1 Score $2 * P * R / P + R$	$2 * 0.25 * 0.75 / (0.25 + 0.75) = 0.37$

2) Logistic Regression Model (LR)

Precision	$60/(160+350) = 0.31$ (31%)
Recall	$60/(160 + 40) = 0.8$ (80%)
F1 Score	$2 * 0.31 * 0.8 / (0.31 + 0.8) = 0.45$

3) Deep Learning Model (DL)

Precision	$80/(180 + 300) = 0.37$ (37%)
Recall	$80/(180+20) = 0.9$ (90%)
F1 Score	$2 * 0.37 * 0.9 / (0.37 + 0.9) = 0.52$

Interpretation:

In terms of both precision and recall, the DL Model performs better than the human and the LR model. For a healthcare organization, higher precision would mean fewer false positives and more accurate prediction of true positives, reducing unnecessary worry and increasing follow-ups and treatments for patients who are incorrectly flagged as negative. A higher recall of 90% means this model accurately captures almost all true positives, which is important for early treatment of diseases where time is of the essence and for preventing missed diagnoses. The implications of missing a false negative for a healthcare organization could be disastrous, not only for its reputation but also for the patient's health outcomes, as the patient could have received treatment promptly had they been flagged correctly. This is why recall is often prioritized in the healthcare industry, as it is considered better to have a few more false alarms than to miss a serious case. In terms of precision, the human model is only 25%, which means only 1 out of 4 cases is flagged correctly. By continuing with this model, the organization risks eroding patient trust and wasting resources on patients who are incorrectly flagged. Although the LR model improves outcomes, the DL model outperforms with a higher F1 Score of 0.53, which is closest to 1, indicating that it best balances catching true positives and avoiding false alarms.

TASK 2: Cost-Benefit Analysis

Cost of False Positives

Human Status Quo Model	$450 * 200 = \$90,000$
LR Model	$350 * 200 = \$70,000$
DL Model	$300 * 200 = \$60,000$

Cost of False Negatives

Human Status Quo Model	$50 * 8000 = \$400,000$
LR Model	$40 * 8000 = \$320,000$
DL Model	$20 * 8000 = \$160,000$

Total Costs

Human Status Quo Model	\$490,000
LR Model	\$390,000
DL Model	\$220,000

Interpretation:

According to these estimates, the DL model reduces the total cost of errors by a large margin, predominantly by reducing the cost of false negatives, which is more serious in this case. Between the human model and the

LR model, the latter reduces costs by \$ 100,000; between the LR and DL models, the costs are further reduced by \$170,000.

Maintaining a uniform rate of false positives at 200 and false negatives at 8000 could simplify the case entirely. As a healthcare organization, the medical staff will be dealing with patients of various demographics and medical complexity. A patient who has stage 3 cancer and is detected as a false positive, and a patient who has a common cold and is detected as a false positive, cannot have a uniform FN cost. The first patient has a faster rate of deterioration and may require complex procedures and surgeries, which will increase the costs. There are also intangible costs such as anxiety, stress, missed work, and litigation costs, which may vary from patient to patient. Similarly, when screen tests rely on imaging such as CT scans, an FP patient can trigger a series of other invasive procedures, such as biopsies, which can raise the FP cost.

TASK 3: Net Tangible Benefit Calculation

$$NTB = (TP * Benefit + TN * Benefit) - (FP * Cost + FN * Cost)$$

Benefit (TP) = \$3000 to \$12,000

Benefit (TN) = \$50 to \$300

FP (Cost) = \$200

FN (Cost) = \$8000

For determining the benefit values, I have chosen three scenarios:

Conservative: For TP = \$3000. For TN = \$50 (to account for minimal admin relief and limited savings)

Base: For TP = \$6500. For TN = 200 (to account for a moderate condition where early detection avoids inpatient stays and reduces patient churn)

Optimistic: For TP = \$12,000. For TN = \$300 (to account for the best-case scenario with large avoided costs value and cost avoidance from clearing TNs)

Human Status Quo Model (TP Benefits)

Conservative	$150 * 3000 = 450,000$
Base	$150 * 6500 = 975,000$
Optimistic	$150 * 12,000 = 1,800,000$

TN Benefits

Conservative	$350 * 50 = 467,500$
Base	$350 * 200 = 1,870,000$
Optimistic	$350 * 300 = 2,805,000$

LR Model (TP Benefits)

Conservative	$160 * 3000 = 480,000$
Base	$160 * 6500 = 1,040,000$
Optimistic	$160 * 12000 = 1,920,000$

TN Benefits

Conservative	$9450 * 50 = 472,500$
Base	$9450 * 200 = 1,890,000$
Optimistic	$9450 * 300 = 2,835,000$

DL Model (TP Benefits)

Conservative	$180 * 3000 = 540,000$
Base	$180 * 6500 = 1,170,000$
Optimistic	$180 * 12,000 = 2,160,000$

TN Benefits

Conservative	$9500 * 50 = 475,000$
Base	$9500 * 200 = 1,900,000$
Optimistic	$9500 * 300 = 2,850,000$

Augmentation Benefits for Human Status Quo Model

Conservative	$(4,50,000 + 4,67,500) - (490,000) = 427,500$
Base	$(9,75,000 + 18,70,000) - (490,000) = 2,355,000$
Optimistic	$(18,00,000 + 28,05,000) - (490,000) = 4,115,000$

Augmentation Benefits for the LR Model

Conservative	$(4,80,000 + 4,72,500) - 390,000 = 562,500$
Base	$(10,40,000 + 18,90,000) - 390,000 = 2,540,000$
Optimistic	$(19,20,000 + 28,35,000) - 390,000 = 4,365,000$

Augmentation Benefits for DL Model

Conservative	$(5,40,000 + 4,75,000) - 220,000 = 795,000$
Base	$(11,70,000 + 19,00,000) - 220,000 = 2,850,000$
Optimistic	$(21,60,000 + 28,50,000) - 220,000 = 4,790,000$

TP/TN benefits vary widely in the real world because of disease severity, and early detection of fast-moving conditions yields huge avoided costs, while slow conditions yield less. Travel, time off work, and insurance status change both TP value and ascertain whether early treatment can occur, and TN savings whether return visits can be avoided.

Automation Benefits for LR and DL Model from the baseline: Human model

LR Model = 2 mins per case plus human confirmation (Note: As the assignment mentions the time + human confirmation, I am assuming that it takes a minute for a human to confirm) (2+1 = 3 minutes total time). However, I understand that this number can vary- increase/decrease depending on the complexity of the case, errors, if any, and the skillset of the staff. For the purposes of doing this analysis, I am assuming that all medical staff take a minute to confirm any case provided to them.

Time saved per task = 15 - 3 = 12 minutes

Task Volume = 10,000 cases

Wage = 40

Automation Benefit = $12 * 10,000 * 40 = 48,00,000$

DL Model = 1 minute per case plus human confirmation (Note: I am assuming that it takes a minute for a human to confirm) (1+1 = 2 minutes total time)

Time saved per task = 15 - 2 = 13 minutes

Task Volume = 10,000 cases

Wage = 40

Automation Benefit = $13 * 10,000 * 40 = 52,00,000$

TASK 4: Comparative Net Tangible Benefit (CNTB)

1) Incremental Benefit gained from the adoption of the LR model compared to the Human Status Quo

Augmentation Benefit: $(\Delta TP * \text{Benefit} + \Delta TN * \text{Benefit}) - (\Delta FP * \text{Cost} + \Delta FN * \text{Cost})$

(Note: I am using the base case scenario of \$6500 and \$200 derived from the previous question for developing CNTB) = $(10 * 6500 + 100 * 200) - (-10 * 8000 + -100 * 200) = 185,000$

Automation Benefit: (Calculated above) = 4,800,000

Total Benefits = 4,985,000

2) Incremental Benefit gained from the adoption of the DL model compared to the LR Model

Augmentation Benefit: $(\Delta TP * \text{Benefit} + \Delta TN * \text{Benefit}) - (\Delta FP * \text{Cost} + \Delta FN * \text{Cost})$

$$= (20 * 6500 + 50 * 200) - (-20 * 8000 + -50 * 200) = 310,000$$

Automation Benefit: LR Model : 3 minutes based on previous assumptions

DL Model: 2 minutes

$$\text{Automation Benefit} = 1 * 10,000 * 40 = 400,000$$

$$\text{Total Benefits} = 710,000$$

Interpretation: The major driver of value for LR is the automation benefit of 4,800,000, which reduces time from 15 minutes to 3 minutes. Most of LR's benefits come from efficiency gains, not accuracy gains, which are relatively small compared to the automation benefit. In contrast, the DL model significantly improves diagnostic accuracy over LR, making it clinically superior. However, its automation benefits compared to LR are fairly small, reducing time per task by only 1 minute. This makes the total incremental benefit smaller than LR's jump from the human model. Value can be interpreted in this case from the perspectives of efficiency (where LR is superior) and clinical accuracy (where DL is superior). The organization can consider initially adopting the LR model, which enables rapid efficiency gains and substantial savings through automation, followed by a transition to DL for long-term sustainability and risk mitigation.

TASK 5: Cost Avoidance vs. Tangible Savings

When calculating NTB, healthcare managers must weigh the value of cost avoidance from true negatives against the more visible True positives. These cost avoidances lead to both tangible (unnecessary diagnostic tests and reduced workload) and intangible (reduced patient anxiety) benefits. Since these benefits are avoided costs and indirect, they are not expressed as revenue. Most NTB models favor visible savings from TP because they are easier to link to and report in financial statements. An approach they could take is to calculate NTB separately for TP and TN and classify them as revenue-driven or savings/operational-driven. Assigning a conservative, modest value to TN will help managers avoid overstating gains.

Modified NTB Formula:

$$\text{NTB} = (\text{TP} * \text{Benefit per TP}) - (\text{FP} * \text{Cost} + \text{FN} * \text{Cost})$$

Assumption: TP Benefit calculated at conservative and optimistic scenarios, per benefit of \$3000 and \$12,000

Human Model:

$$(150 * 3000) - (450 * 200 + 50 * 8000) = -40,000$$

$$(150 * 12,000) - (450 * 200 + 50 * 8000) = 1,310,000$$

LR Model:

$$(160 * 3000) - (350 * 200 + 40 * 8000) = 90,000$$

$$(160 * 12,000) - (350 * 200 + 40 * 8000) = 1,530,000$$

DL Model:

$$(180 * 3000) - (300 * 200 + 20 * 8000) = 320,000$$

$$(180 * 12,000) - (300 * 200 + 20 * 8000) = 1,940,000$$

Even after excluding TN benefit, the DL model remains the clear choice, focusing only on clinical impact and cost savings. However, TN should not be completely dismissed, as it has the potential to reduce resource wastage and staff workload. A modified NTB with a conservative estimate can be used for budgeting decisions when the organization is not well-off financially. Based on the development and integration costs, the organization can account for that and accordingly pick the LR or DL model. The human model should be dropped, as it results in losses. In a high-performing organization, utilizing original NTB with TP and TN benefits can be justified in conservative and base-case scenarios when discussing broader benefits for patients and staff.

TASK 6: Scenario-Based Analysis and Long-Term Impact**Scenario A: Prioritizing Patient Experience and Minimizing Anxiety**

The goal in this scenario is to reduce unnecessary follow-ups and tests and to build the patient's trust in the screening process. This means reducing the rate of false positives, and in such a scenario, placing a heavier penalty or cost on this rate would be beneficial. Each patient incorrectly flagged may have to undergo unnecessary procedures and costly tests.

A modified NTB that accounts for this scenario, with a higher cost assigned to FP and a slight reduction in costs assigned to FN, would best reflect the organization's priorities.

Current Costs under FP and FN = \$200 AND \$8000

A 50% markup under FP and a 10% markdown in FN costs (Assumption: To place a higher cost, I am assuming the organization can increase the cost of FP and reduce the cost of FN based on these estimates. I understand that another approach is to assign penalties/weights; however, I followed this because no specific approach was explicitly mentioned in the assignment.

New FP = \$300

New FN = \$7,200

Assumption: base case scenario of \$6500 per TP and a conservative estimate of TN = \$50

Human Model:

$$\text{Modified NTB} = (\text{TP} * \text{Benefit} + \text{TN} * \text{Benefit}) - (\text{FP} * \text{Cost} + \text{FN} * \text{Cost}) =$$

$$(150 * 6500 + 9,350 * 50) - (450 * 300 + 50 * 7200) = 947,500$$

LR Model:

$$\text{Modified NTB} = (160 * 6500 + 9450 * 50) - (350 * 300 + 40 * 7200) = 1,119,500$$

DL Model:

$$\text{Modified NTB} = (180 * 6500 + 9500 * 50) - (300 * 300 + 20 * 7200) = 1,411,000$$

Since the DL model reduces FP by an incremental 150 relative to the human status quo and by 50 relative to the LR model, and it also has fewer FNs, it performs better than the LR and human models. Raising the cost of FP to \$300 hurts high-FP methods, but since DL has the lowest count, it's the least affected. Similarly, trimming FN costs to 7200 reduces overall NTB across all scenarios, but DL continues to dominate because it avoids most FN.

Long-term sustainability: As the FN cost has been deprioritized, organizations have a lower incentive to invest in highly sensitive diagnostic models like the DL, even though it yields the largest NTB, not accounting for implementation costs. Long-term, whether the DL or LR model is picked, depending on the organization's priorities, there will be a need to retrain the model to improve specificity and reduce false positives. Once access to higher-quality data is provided, both precision and recall improve, and these improvements can protect the organization against public scrutiny of fairness in these models and against regulatory changes that mandate higher sensitivity. High FP volumes overload testing systems, delaying the care for TP patients. Pairing these models with workflows that prioritize high-risk patients will help mitigate these consequences.

Scenario 2: Prioritizing diagnostic sensitivity over patient experience

In this scenario, prioritizing the detection of almost every positive case is imperative, even if some patients are incorrectly flagged. This scenario becomes important in situations where missing a diagnosis can be life-threatening or alter the healthcare industry. In this case, reducing FN and placing higher penalties on this rate, while reducing emphasis on FP, would best reflect the organization's priorities.

Assuming a 50% markup under FN and a 10% markdown in FP costs

New FN = 12,000

New FP = 180

Assumption-based case scenario of \$6500 per TP and a conservative estimate of TN = \$50

Human Model:

$$\text{Modified NTB} = (\text{TP} * \text{Benefit} + \text{TN} * \text{Benefit}) - (\text{FP} * \text{Cost} + \text{FN} * \text{Cost}) = (150 * 6500 + 9,350 * 50) - (450 * 180 + 50 * 12,000) = 761,500$$

LR Model:

$$\text{Modified NTB} = (160 * 6500 + 9450 * 50) - (350 * 180 + 40 * 12,000) = 969,500$$

DL Model:

$$\text{Modified NTB} = (180 * 6500 + 9500 * 50) - (300 * 180 + 20 * 12,000) = 1,351,000$$

In this scenario as well, the DL model is the superior choice, having the highest recall, that is, catching almost all true positives and reducing false negative cases massively. The reduced FN count under the DL model yields greater benefits. This aligns with the organization's priorities in high-stakes disease-detection environments, such as cancer and pandemics.

Long-term sustainability: As the model continues to improve, FN rates may decrease overall, increasing the NTB. As data continues to expand in the model and population changes occur, the model will need to be retrained and tested. As the DL model is retrained with more real-world data, its ability to correctly identify cases will improve. With a trend towards high-sensitivity training, the DL model appears better aligned with healthcare industry standards and regulations. Regular bias audits and fairness checks will also need to be implemented in a timely manner to ensure that subgroups in the population are equally protected. Retraining the model with more diverse datasets will ensure equitable accuracy and build long-term trust in the system. As the healthcare landscape evolves rapidly with an influx of new diagnostic tools and diseases, the model should be robust yet flexible enough to incorporate new disease information. This process of data growth, retraining, and system upgrades will require close collaboration between data scientists, medical staff, and software engineers. It will also involve examining long-term fixed costs, such as development, integration, and operations, which are not included in the current study.

Value Measurement Framework

- Step 1: Setting goals, which includes picking which scenario the organization is going with - patient experience, sensitivity, or looking at both equally. This step would also include setting cost-benefit baselines and locking in assumptions about the benefits and costs associated with correct and incorrect flagging.
- Step 2: Defining the unit of analysis and calculating the NTB and CNTB for each scenario with the model. The unit could be a screening of one patient, and the cases selected for the calculations should include key subgroups such as age, sex, gender, ethnicity, race, insurance, and so on. The calculations should be performed for the overall NTB and for a modified NTB with only TP to prevent bias towards efficiency.
- Step 3: Include data sources and establish event timelines that show: case -> AI Model test -> human review time -> follow-up tests -> final diagnosis -> treatment. Conduct data quality tests here for completeness, accuracy, and to remove duplicates.
- Step 4: Setting up governance and fairness in testing, including roles and responsibilities across a cross-functional team, with monthly reviews and quarterly value delivered by the AI model.
- Step 5: Pilot the design across sites with the pre-specified criteria
- Step 6: Defining the KPI dashboard. This includes looking at Clinical criteria such as sensitivity and specificity, operational criteria such as time saved per task, average time per case, etc., and financial criteria, including augmentation ROI, taking into account development and integration costs, NTB (standard and No-TN), and labor dollars saved. Acknowledge both soft and hard benefits of the AI model adoption.
- Step 7: Establish a continuous learning and retraining plan with feedback loops between groups.

To ensure that the system is sustainable and operational in the long run, it is imperative to monitor value delivery across metrics continuously:

Precision (Example: FP count)

Patient experience metrics (Example: volume of complaints, surveys on satisfaction)

Financial Metrics (Example: NTB and ROI)

Operational Metrics (Example: Workflow delays, staffing overload)

Equity Metrics (Looking at correct flagging across different demographic subgroups)

APPENDIX ->

Used CHAT GPT 5 THINKING

Q) What metrics should I include for a value measurement framework for an AI model that is being used in the healthcare industry? (Concise answer, only give me metrics, no calculations or examples)

ANS:

NTB

FN Volume

FP Volume

Set goals and primary aim

Lock the value formulas

Governance and RACI: Value/ROI owner: finance + service-line lead.

- Clinical safety & equity: CMO + quality + patient rep.
- Tech/MLOps: data science + IT.
- Cadence: monthly ops review; quarterly value & equity board; annual re-authorization. Threshold tuning & cost-sensitive optimization
- Quarterly, refit the operating point to minimize your current cost function. Consider subgroup-specific thresholds if allowed and governed, to reduce equity gaps.

Q) For this question, should I use the total number of cases in each model or compare between models to see the incremental benefit for augmentation? Similarly, for augmentation, I would look at the incremental gain in time saved by using the AI model. (Concise answer please)

ANS: Use total numbers for each model when calculating NTB, since NTB represents the absolute value generated by that model (accuracy + efficiency).

However, for augmentation benefit or incremental gains, compare between models:

- Augmentation (accuracy): Calculate the *difference* in TP, TN, FP, and FN between models (e.g., AI vs. Human) to show improvement.

Q) I am a bit confused about how to adjust the FP and FN costs in Scenario A and Scenario B. If I increase the cost of FPs by 50% and reduce FN costs by 10%, should I apply those changes only to the modified NTB formula, or should I also recalculate my base case values? I just want to make sure my comparisons remain consistent across scenarios.

ANS:

You only apply the **adjusted FP and FN costs to the modified NTB formula** for the scenario you are evaluating.

- The **base-case values should remain constant to provide** a consistent point of comparison across all models and scenarios.
- This allows you to isolate the effects of changing organizational priorities (e.g., patient experience vs. diagnostic sensitivity) without distorting the baseline data.
- In practice, think of the base case as your “control,” and the modified FP/FN costs as scenario-specific “weights” that reflect what matters most in that scenario.
- DL might not be chosen because its higher recall comes at the cost of more FPs, undermining patient experience.

NOTE: For the scenario-based one, I did not follow ChatGPT’s advice, as it suggested that the LR model or the Human model would be better when the weights are placed, which is not the approach I took. I adopted a simplistic approach that increased or decreased costs by a certain percentage to reflect the organization’s priorities. Based on my analysis, it was the DL model that was the superior choice (In ChatGPT’s model, the FN is completely de-emphasized; I did not de-prioritize it completely, as for any health care organization, avoiding FN’s or reducing it is extremely important to protect patients and begin treatment early for those flagged correctly). It is also important for them to protect their reputation and avoid litigation.